

System Design Specifications

System Overview:

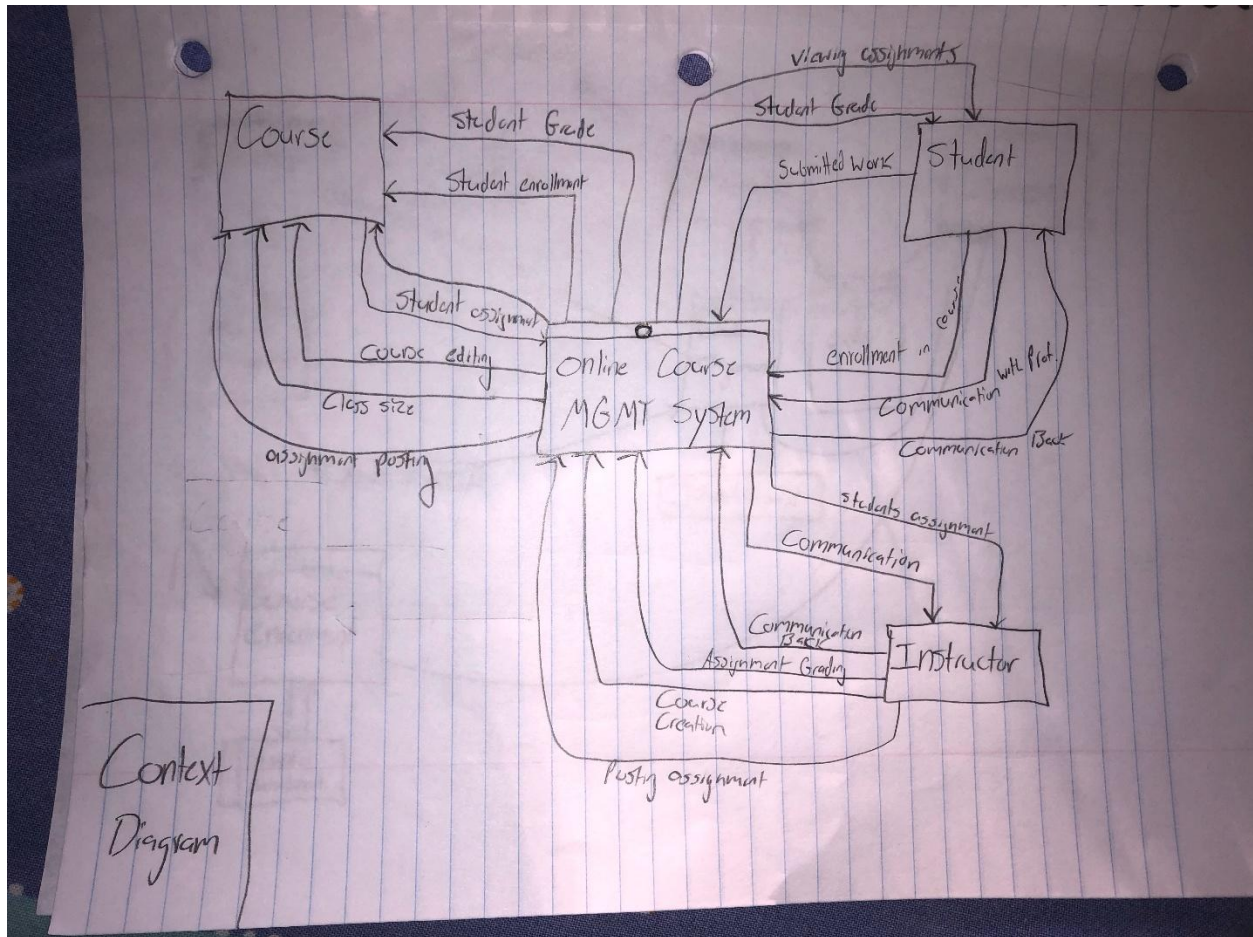
The online management system for a university will be an all online all-in-one website that will allow students to enroll in and drop classes, submit assignments for grading, and allow instructors the ability to create courses of their own design, grade student submissions, and will allow communication between both student and their instructor.

-Goals/Objectives:

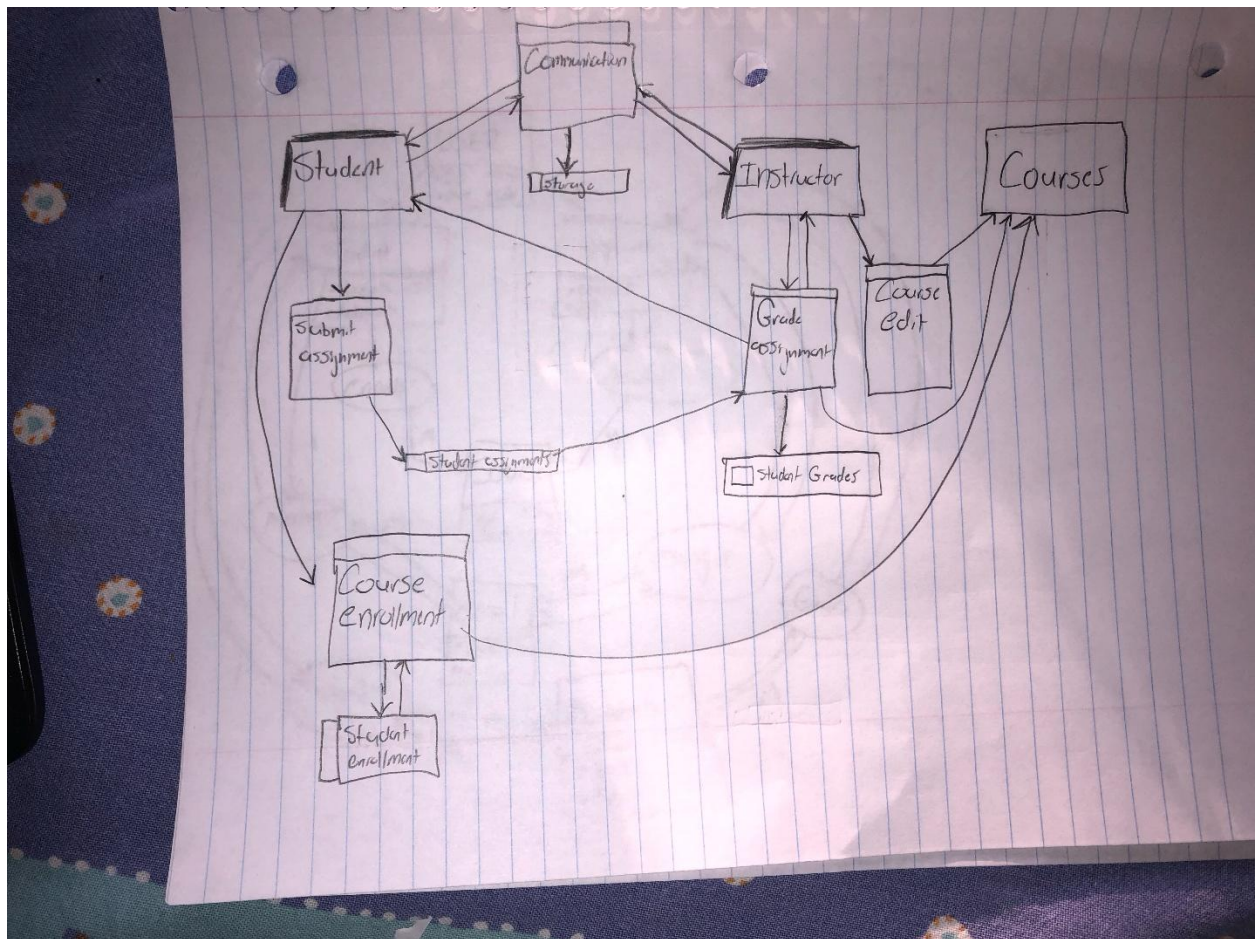
- All in one course creation and management into one system
- Student enrollment and course drop ability
- Instructor course creation and editing
- Submission of assignments, grading, and communication between student and instructor.
- Secure storage of communications and grades of students and Instructors
- Secure login with different privileges for both students and faculty

Design Integration

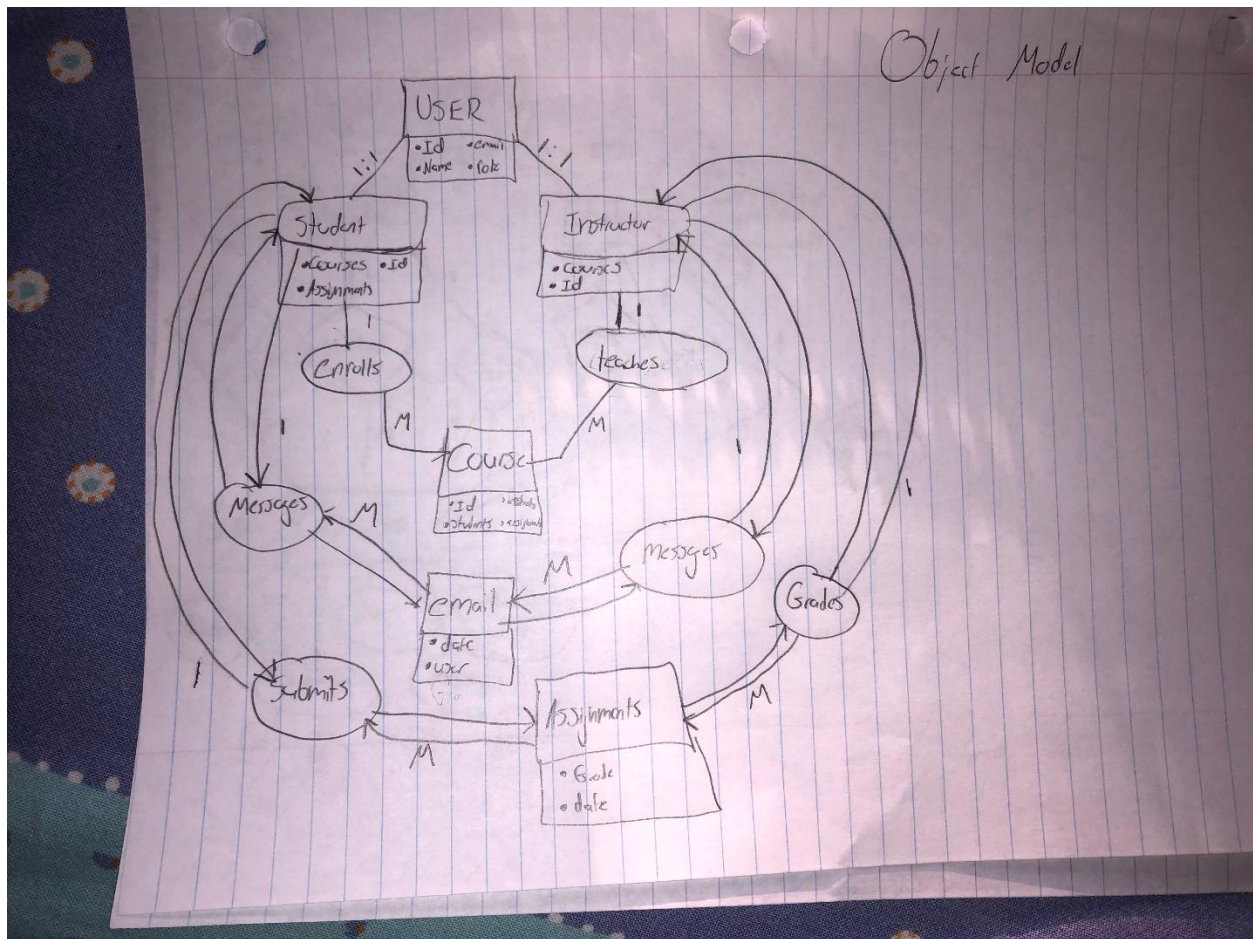
-Context Diagram:



-Data Flow Diagram (DFD):

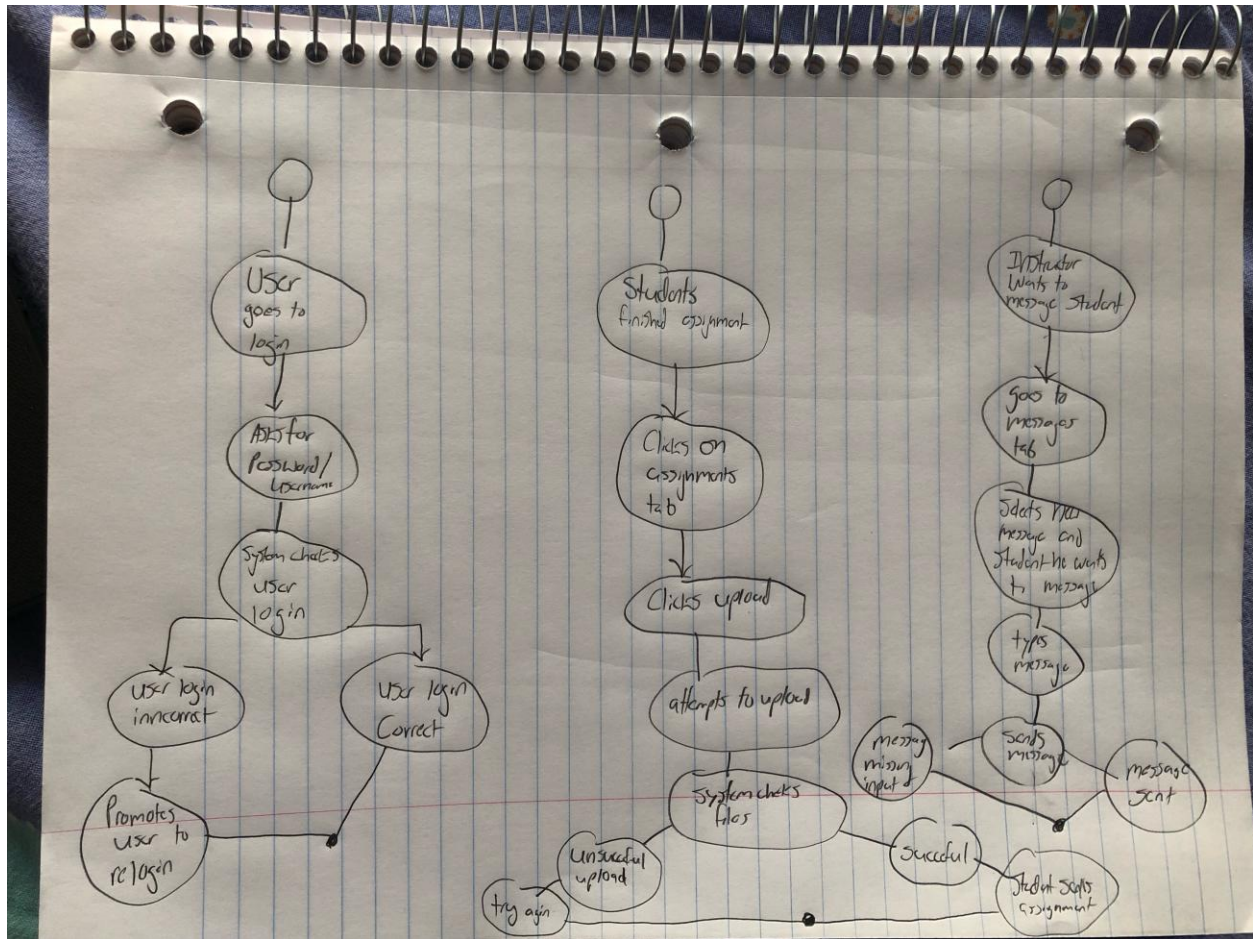


-Object Model (Class Diagram):

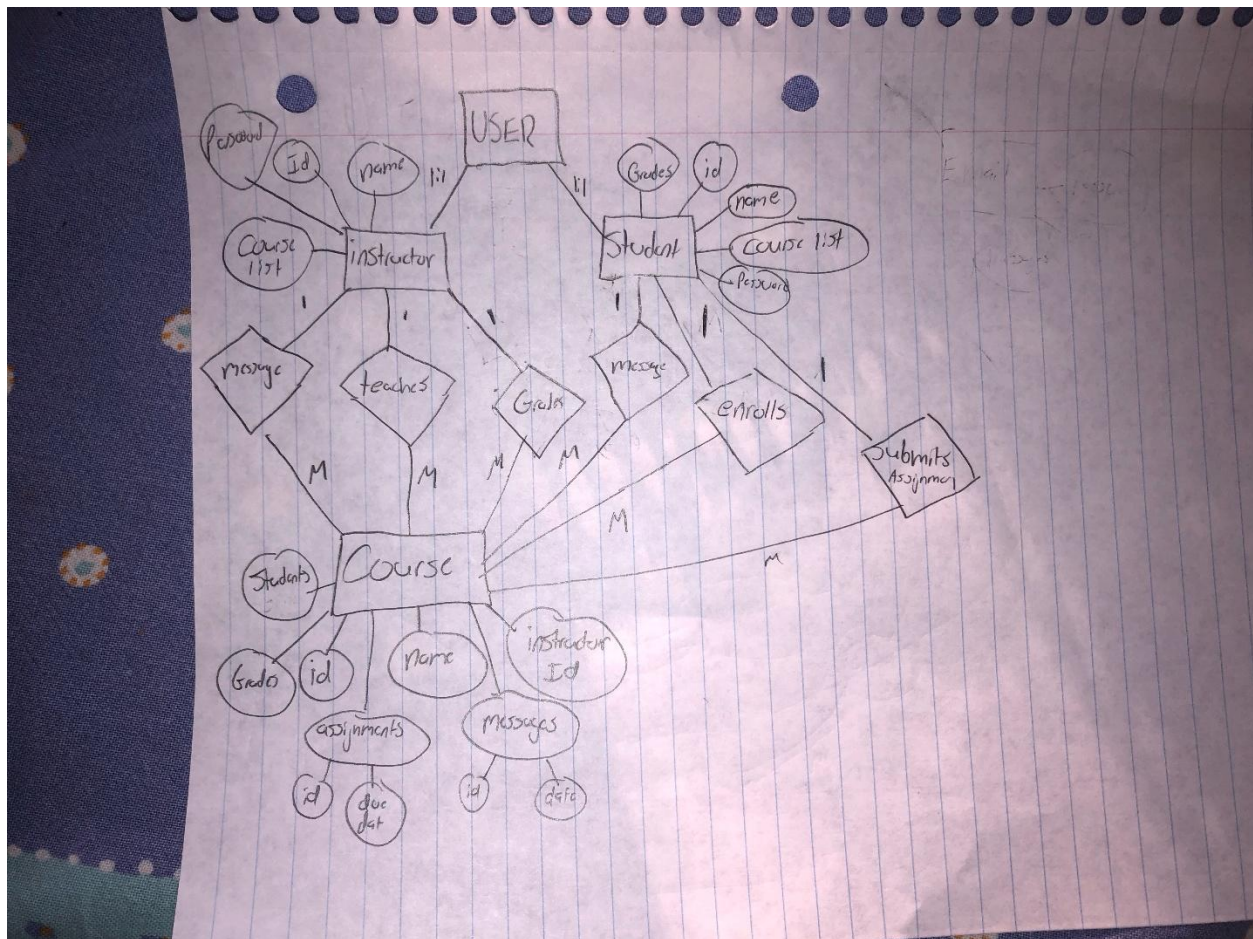


-Use Cases:

- **User Login:** Goes to website, login button, types in password and username, clicks login, systems checks user login information, either denies access and prompts user to reenter information or user logged in and continues to homepage.
- **Student Assignment submit:** Student done with assignment, logs in to website, goes to assignments page, finds assignments, clicks upload file, attempts to upload file, either upload successful or incompatible file upload and prompted to redo. If successful, submit file.
- **Instructor wishes to message student,** goes to messages tabs, enters who they wish to message and compose message, hit send and either the message goes out to the student or the systems finds an error in the inputs and prompts user to review.

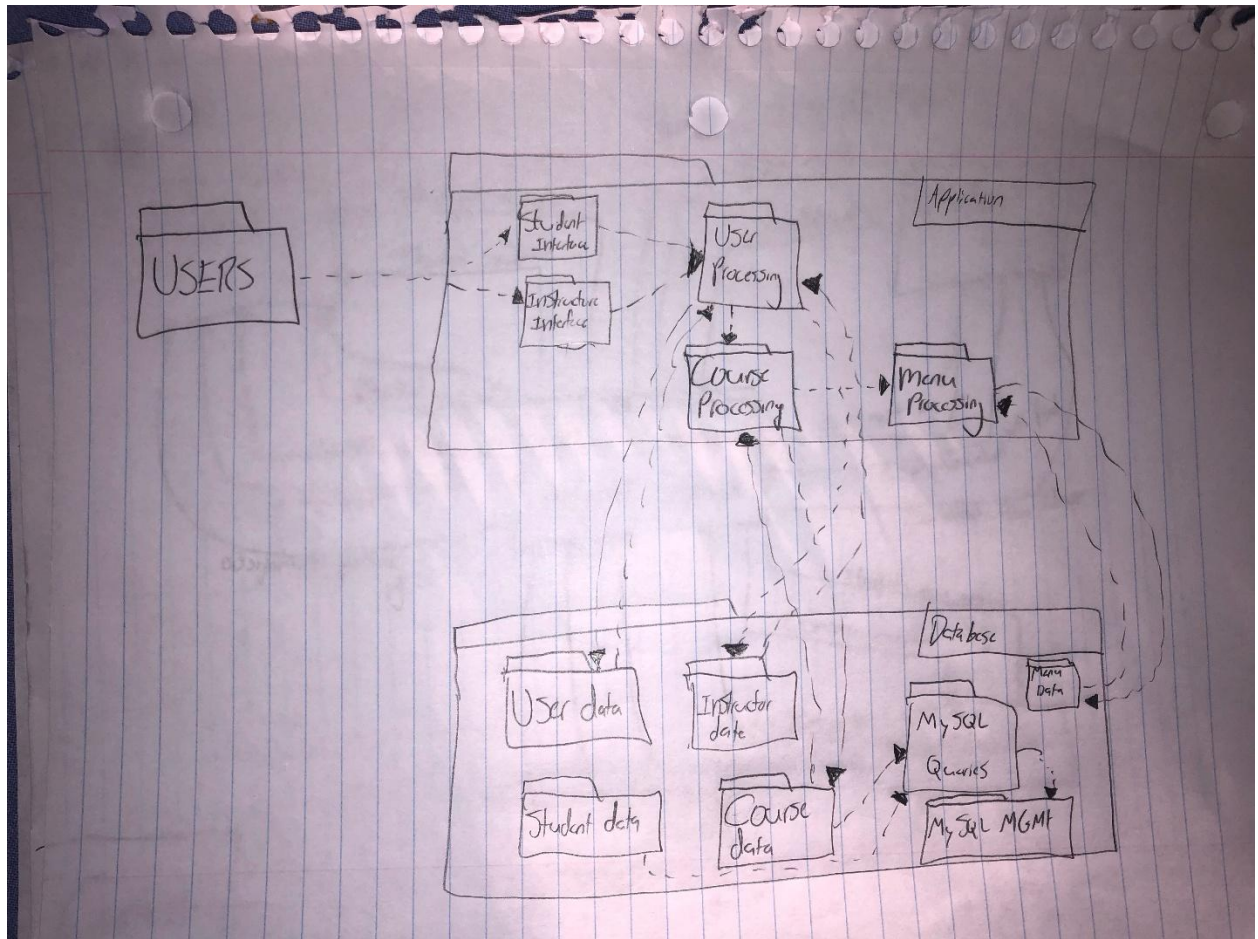


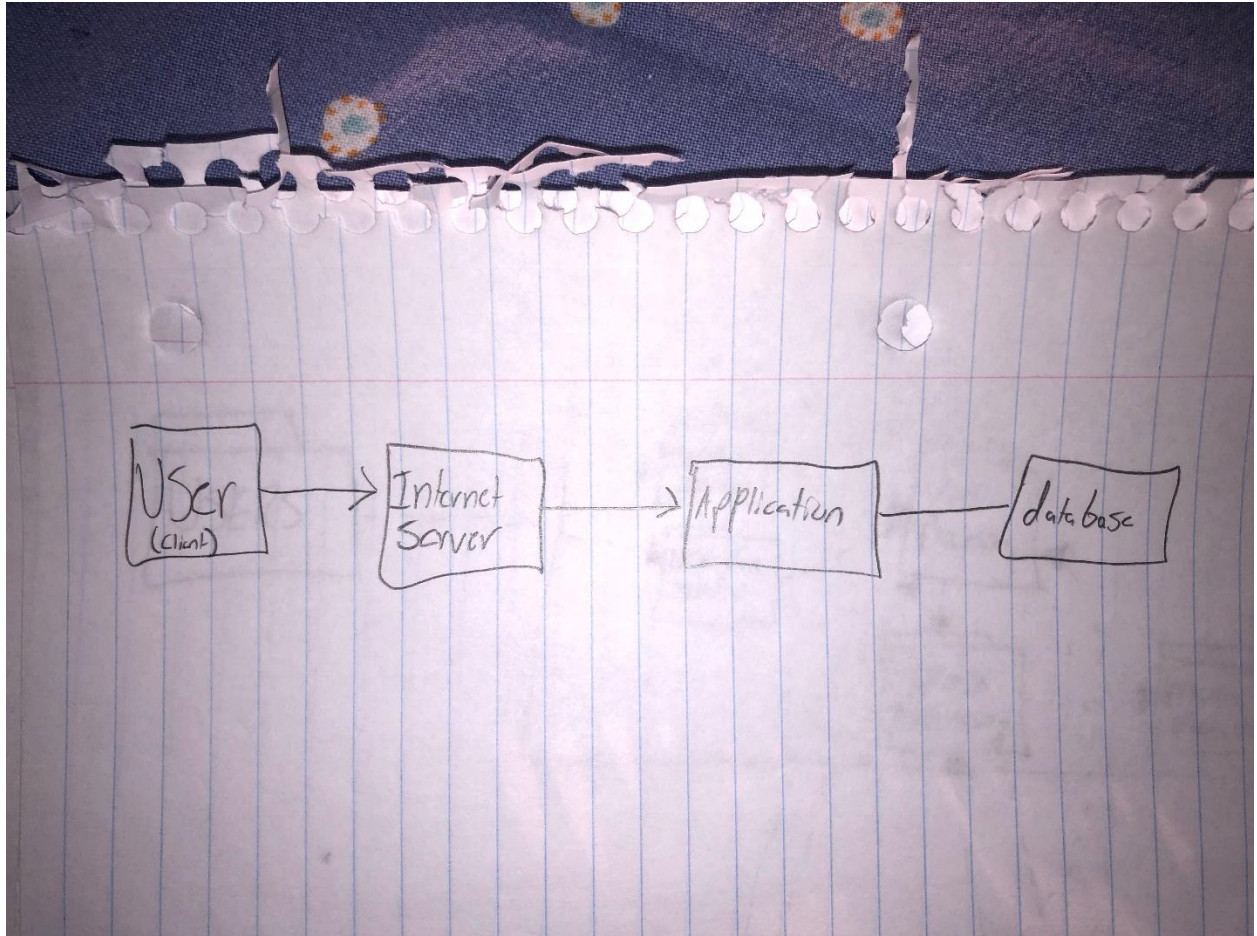
-ERD:



Architecture and UI

-System Architecture Diagram:

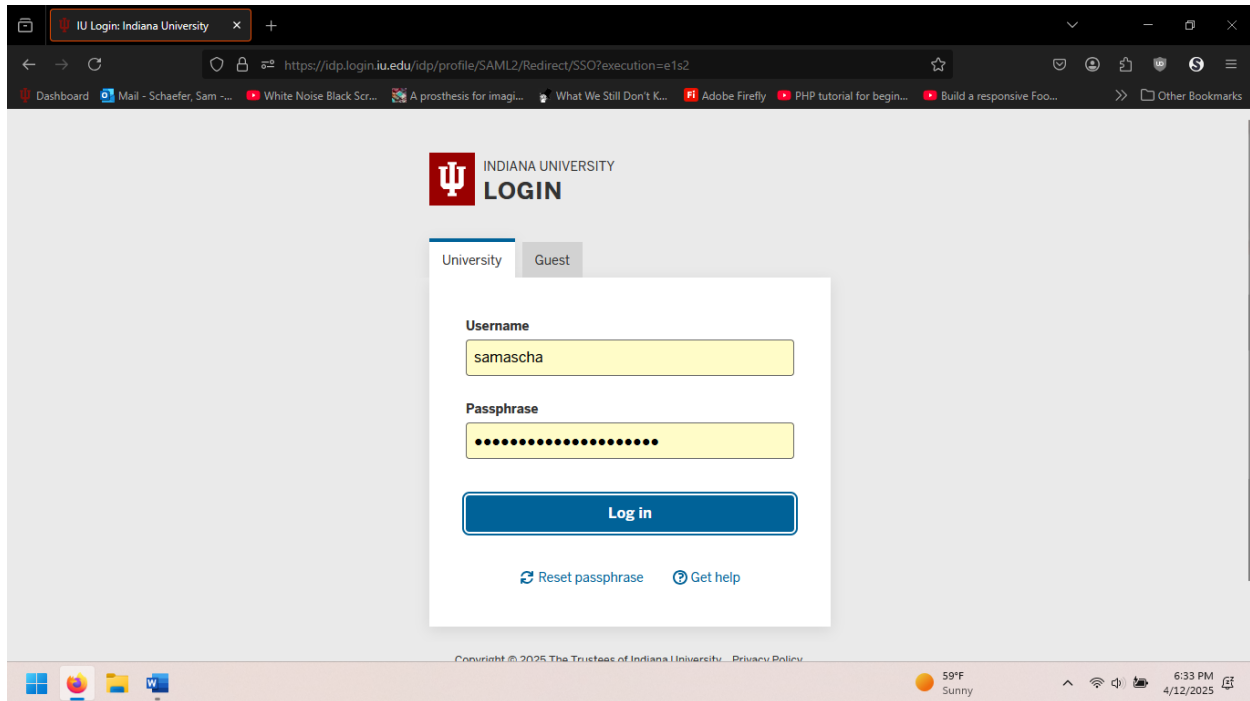




Sample UI Mockups: (Sampled template from IUS.EDU)

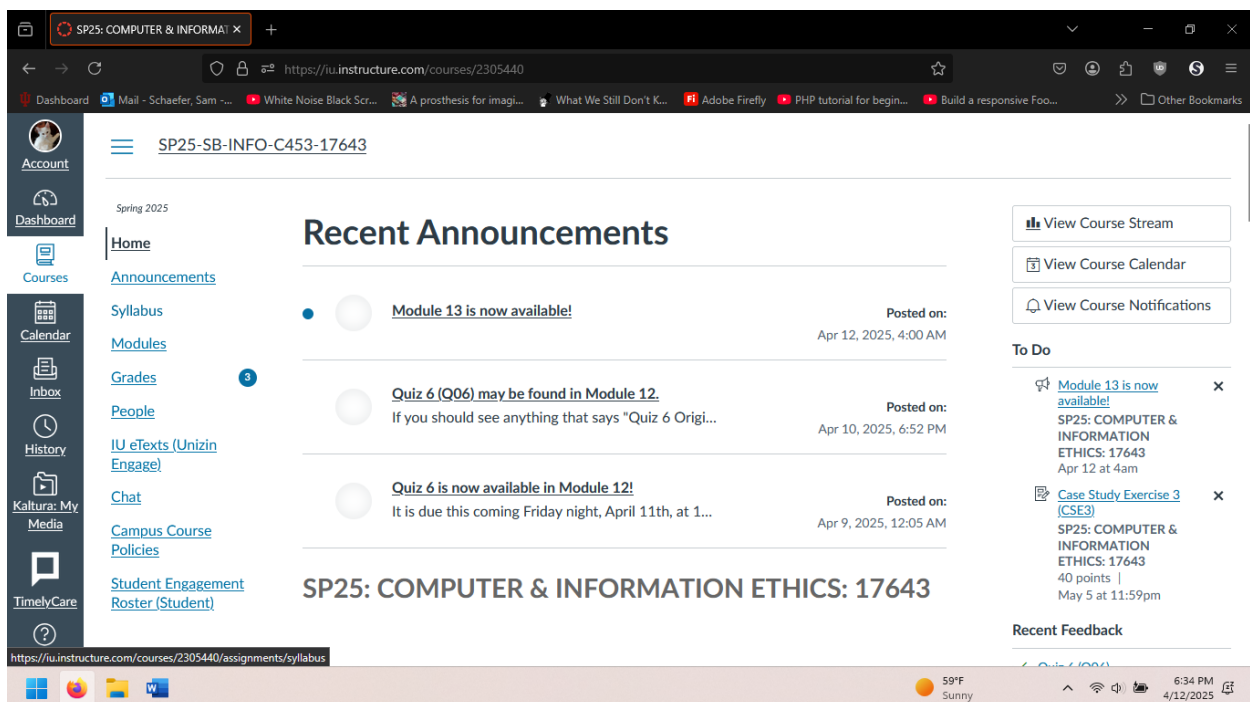
-Login:

- Username, password, login button.



-Student Dashboard:

- courses, announcements, assignments due, grades, messages.



-Assignment Submission Page:

- Upload assignments, due dates, points

Assignments: SP25- WEB DESIG

SP25-SB-INFO-C413-17771 > Assignments

Spring 2025

Search...

SHOW BY DATE SHOW BY TYPE

Upcoming Assignments

- Assign5**
Available until Apr 13 at 11:59pm | Due Apr 13 at 11:59pm | ~20 pts
- Week 14-15 Your Design**
Not available until Apr 14 at 12am | Due May 4 at 11:59pm | ~15 pts
- Week 14-15 Feedback**
Not available until Apr 21 at 12am | Due May 4 at 11:59pm | ~5 pts
- Final Exam (Using Proctoring)**
Not available until May 5 at 12am | Due May 8 at 11:59pm | ~100 pts

Past Assignments

Assign5

Due: Sun Apr 13, 2025 11:59pm

20 Points Possible

Attempt 1

In Progress
NEXT UP: Submit Assignment

Add Comment

Unlimited Attempts Allowed
Available: Mar 27, 2025 12:00am until Apr 13, 2025 11:59pm

Program Assessment

This assignment is being used to assess one or more IU Online (IUO) program learning outcome(s). The rubric for this assignment includes rows specifically addressing these learning outcomes. Please note: The IUO program criteria do not contribute to the final score on this assignment, and they will not impact your course grade.

Submit Assignment

Navigation Flow and Usability Considerations

- User dependent dashboard (student or Instructor)

- Tabs for Courses, Assignments, Grades, messages
- Arrow buttons to go back

Security and Risk Considerations

Security Features:

- User dependent access
- Encrypted messages and data
- Data Validation

Risks and Mitigation Strategies:

-Risks:

- 1.) Data hacks or breeches
- 2.) Data disruption or loss
- 3.) Service constraints

-Mitigation:

- Data encryption and user authentication processes
- Backups storage of user data
- Backup servers and load management

Design Patterns & Best Practices

-Design Patterns Used:

- Observer: object maintains list of its dependents, the observer, and notifies them when any changes take place by calling a method.
 - Rational: allows students to see any announcements, grade posting, and new assignments and vice versa for the instructors if they get a message from a student.
- Adaptor Method: allows the interface of an existing class to be used as another interface.

- Rational: Allows you to change or update other parts of code if you change certain parts like a messaging service or different grading methods and this pattern helps you adapt the code to work within your already in place code, so you don't have to go around and change everything.